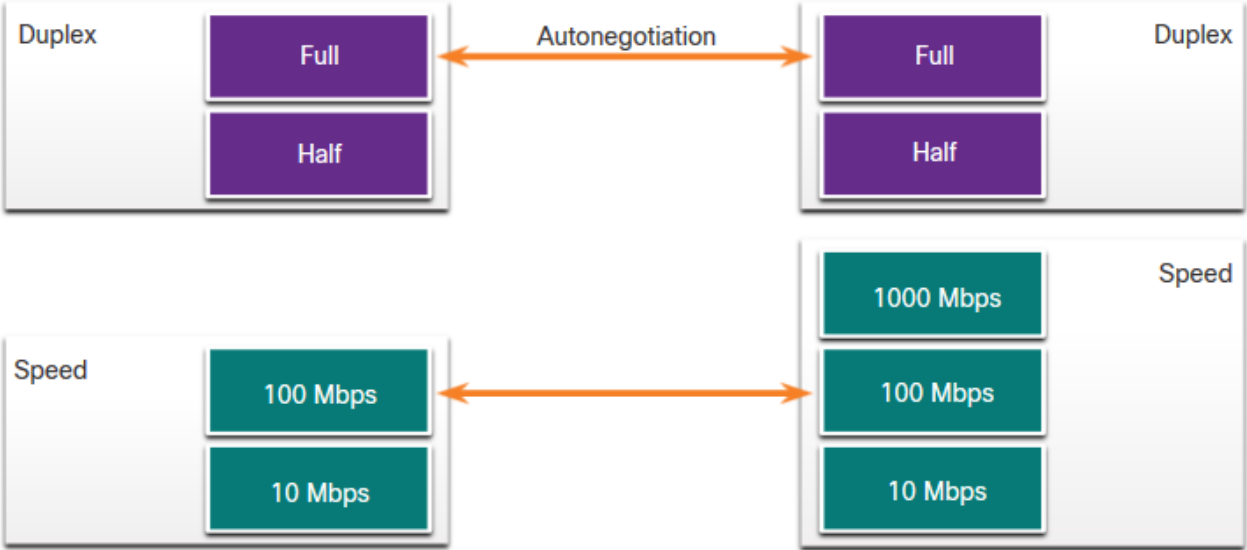


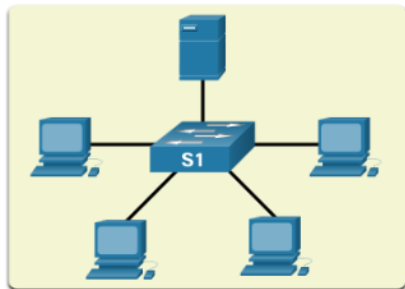
2.2 Switching Domains

Switching Domains

Collision Domains

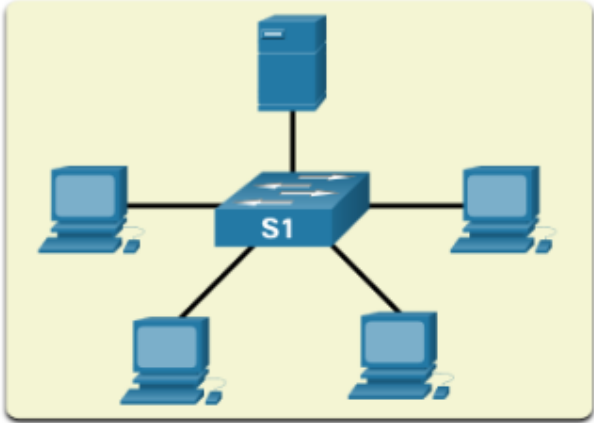


Broadcast Domains



- A broadcast domain extends across all Layer 1 or Layer 2 devices on a LAN.
- Only a layer 3 device (router) will break the broadcast domain, also called a MAC broadcast domain.
- The broadcast domain consists of all devices on the LAN that receive the broadcast traffic.
- When the layer 2 switch receives the broadcast it will flood it out all interfaces except for the ingress interface.
- Too many broadcasts may cause congestion and poor network performance.
- Increasing devices at Layer 1 or layer 2 will cause the broadcast domain to expand.

Broadcast Domains



Alleviated Network Congestion

Switches use the MAC address table and full-duplex to eliminate collisions and avoid congestion.

Features of the switch that alleviate congestion are as follows:

Protocol	Function
Fast Port Speeds	Depending on the model, switches may have up to 100Gbps port speeds.
Fast Internal Switching	This uses fast internal bus or shared memory to improve performance.
Large Frame Buffers	This allows for temporary storage while processing large quantities of frames.
High Port Density	This provides many ports for devices to be connected to LAN with less cost. This also provides for more local traffic with less congestion.

2.3 Module Practice and Quiz

What did I learn in this module?

Frame Forwarding

- Ingress is the entry port, egress is the exit port.
- The switch builds a MAC address table to forward frames on the LAN.
- The switch can use either the store-and-forward or cut-through method of switch forwarding.

Switching Domains

- Ethernet ports in half-duplex will be a part of a collision domain.
- Full-duplex will eliminate collision domains.
- A switch will flood out all interfaces except the ingress port if the frame is a broadcast or if the unicast destination MAC is unknown.
- Broadcast domains may be broken up by a layer 3 device, like a router.
- Switches extend broadcast domains, but can eliminate collision domains and relieve congestion.

